MILITARY SPECIFICATION

SYRINGE, HYPODERMIC, DISPOSABLE, ALLERGY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a sterile, disposable, plastic syringe with a hubless, nondetachable intradermal needle.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-S-36586	Sterilization Test Strip Set, Bacterial Spore.

STANDARDS

Federal

FED-STD-151	Metal, Test Methods Nos. 111 and 112.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:54Z
Check the source to verify that this is the current version before use.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

FEDERAL FOOD AND DRUG ADMINISTRATION

Federal, Food, Drug and Cosmetic Act and regulations promulgated thereunder.

(Application for copies should be addressed to the Food and Drug Administration, 5600 Fishers Lane, Rockville, Maryland 20852.)

U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

3. REQUIREMENTS

3.1 Material. All material shall comply with the requirements of the Federal Food, Drug, and Cosmetic Act. The portions of the unit that come in contact with the fluid to be injected shall be compatible with all known hypodermic medications.

3.1.2 Barrel. Syringe barrels shall be fabricated from rigid, transparent, virgin, nontoxic, high-grade polypropylene plastic (see 3.1). The barrel shall be considered to be transparent if the scale can be seen in daylight by the naked eye when viewed through the empty barrel from the side opposite to the scale.

3.1.2.1 Alcohol resistance of graduation markings. The graduation markings shall withstand vigorous rubbing of an alcohol moistened cotton swab and remain legible when tested according to 4.4.6.

3.1.3 Plunger. The plunger rod shall be fabricated from rigid polypropylene plastic. The plunger tip shall be fabricated from fresh stock, pharmaceutical grade, nonblooming, nonleaching, nontoxic, age-resistant, inert, natural rubber, or neoprene (see 3.1).

3.1.3.1 Inertness to medication. The plunger tip shall show no evidence of a reaction with a high pH buffer solution when tested according to 4.4.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:54Z
Check the source to verify that this is the current version before use.

3.1.4 Cannula. The cannula shall be fabricated from either seamless drawn or welded and drawn stainless steel tubing AISI Type 304. Material shall be tested as specified in 4.4.8 for chemical composition.

3.1.5 Needle protectors. Each syringe shall be provided with a protector fabricated from a suitable rigid, non-toxic, high grade plastic.

3.1.6 Lubricant. Shall be an inert, non-toxic silicone compound suitable for lubricating the movement of the plunger in the barrel. The silicone compound shall be an FDA approved type.

3.2 Style, design, and construction. Shall be a sterile, pyrogen-free, non-toxic, disposable syringe with a hubless, non-detachable intradermal needle suitable for a single administration. Each syringe shall be supplied with a needle protector to maintain sterility.

3.2.1 Barrel and needle. The syringe barrel shall be fabricated from plastic, as specified in 3.1.2. The barrel shall consist of a tube closed at one end except for an orifice and open at the other end to receive the plunger. At the closed end there shall be a permanently affixed 26 gage needle of 3/8 inch length. The needle shall be located concentrically with respect to the long axis of the barrel. The wall at the open end of the barrel shall be extended radially to form a flange of suitable size and shape to provide finger grip and anti-roll features. The inside of the barrel shall have a very fine, uniform surface, free from pits, grind marks, bubbles, mold marks, flash, open air lines, or other defects that would affect the fluid tightness and smoothness of its fit with the plunger. The inside surface of the barrel shall be lightly coated with the silicone lubricant specified in 3.1.6. Dimensions of the barrel shall be as specified in Table I. Inside diameter shall be such as provide an accurate fit with plunger tip, resulting in a fluid-tight fit (see 3.2.2 and 3.4), and yet permitting plunger to move freely throughout its excursion in the barrel. Ungraduated portion of barrel, at proximal end, shall be of sufficient length to permit plunger tip to be aligned with top graduation mark, while barrel is encompassing entire rubber tip, but shall be not greater than one-third of the overall length of the syringe barrel.

Table I. Barrel Dimensions and Graduations

Size	Overall length*	Outside diameter**	Wall thickness	Subgraduations	Main graduations	Numbering
1 (1cc)	2-1/2 to 4	0.2 to 0.4	0.020 (minimum)	0.01 cc	0.05 cc	0.1 cc

Values in inches unless otherwise stated.

* Excluding needle.
** Measured near distal end.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:54Z
Check the source to verify that this is the current version before use.

3.2.1.1 Graduations. The outside surfaces of the barrel shall be marked with a graduation scale and the name or registered trademark of the contractor. The graduation scale shall be applied to the barrel in such a manner as to be approximately in line with the flattened side of the flange. The zero (0) mark of the scale shall register precisely with the distal ring of the plunger tip, when plunger is pushed to the bottom of the barrel. The scale and numbering shall be in cubic centimeters as specified in Table I. Main graduation marks shall be at least 33-1/3 percent longer than subgraduation marks. No number is required at the zero (0) mark. The markings shall be legible and of a dark contrasting color, easily readable. The unit "cc" shall be clearly marked immediately above the highest graduation mark.

3.2.2 Plunger. The plunger shall be either hollow or solid, fabricated from the material specified in 3.1.3. The distal end of the plunger shall be shaped to permit attachment of a plunger tip. The proximal end of the plunger shall extend well out of the barrel, and shall be shaped to form a knob of a thickness and diameter appropriate to the size of the syringe, and shaped to allow a suitable finger grip for usage. The distal end of each plunger with its attached plunger tip shall accurately fit its individual barrel so as to make the syringe successfully withstand the fluid tightness requirements specified in 3.4, permit the plunger and plunger tip to draw liquids, and yet move freely and smoothly throughout its excursion in the barrel. The plunger tip shall be securely attached to the plunger rod so as not to become loose or detached when in use. The tip shall be shaped to closely fit the distal end of the barrel so as to remove all fluid from the inside of the barrel when the plunger is pushed all the way in. In addition, the tip shall be designed, e.g., beveled, in order to permit easy and efficient removal of air bubbles from fluid path. The plunger tip shall be properly treated with the silicone lubricant specified in 3.1.6. An excess of lubricant that detracts from appearance, interferes with visibility of plunger tip ring, or can be caused to form pools or droplets, or can be washed out with the solution being used, shall be cause for rejection. Graduation markings, denoting correct, accurate capacity, shall register precisely with the first ring at the plunger tip distal end, touching sides of the barrel. Tip shall be of a permanent, suitable color so that the first ring at the plunger tip distal end, touching sides of the barrel, shall be clearly visible and shall accurately indicate through the barrel wall. Thumbrest shall be perpendicular to the centerline of the plunger and may be flat or slightly curved. As an alternate, thumbrest may have a centered hole which shall not exceed 3/8 inch in diameter. Thumbrest shall be fabricated from material as specified in 3.1.3.

3.2.3 Needle. The needle shall have the dimensional characteristics of a 26 gage needle and 3/8-inch length with an intradermal point of very short bevel length. The dimensions in inches shall be as listed in Table II.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:54Z
Check the source to verify that this is the currentversion before use.

Table II. Dimensions of Needle

Gage	Point length*	Length	Outside diameter	Inside diameter at point, minimum
26	0.04 ± 10%	3/8 ± 3/64	0.175 to 0.185	0.0095 min

* Measured from heel to tip and parallel to longitudinal axis of cannula.

3.2.3.1 Corrosion-resistance of cannula. When tested as specified in 4.4.13, the cannula shall show no appreciable corrosion.

3.2.3.2 Security of assembly. When tested as specified in 4.4.10, the cannula shall not loosen or pull out of the barrel.

3.2.3.3 Resistance to breakage. When tested as specified in 4.4.11, there shall be no breakage.

3.2.3.4 Stiffness. When tested as specified in 4.4.9, the maximum deflection shall not exceed 0.019 inch.

3.2.4 Needle protector. Each syringe shall be provided with a protector. The protector shall fit securely over the syringe tip, and needle covering the needle cannula, and shall maintain the sterility of the needle and syringe tip. The protector shall be so secured on the syringe as to remain in place during shipment and normal handling. Needle protectors without hermetically sealed ends shall be adequately protected with cotton or a paper seal to maintain the required sterility.

3.3 Accuracy. When tested as specified in 4.4.1, syringes shall deliver plus or minus 5 percent of the capacity indicated at each graduation mark tested.

3.4 Quality of seal between plunger and barrel. The syringe shall be tested for quality of seal between the plunger and barrel, as specified in 4.4.2. There shall be no air leakage as revealed by the suction test as specified in 4.4.2.

3.5 Sterility. The syringe shall be sterile when tested with a biological indicator as specified in 4.4.5.

3.5.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

Source: https://assist.dla.mil -- Downloaded: 20260-21T09:54Z
Check the source to verify that this is the current version before use.

a. Item identification.
b. National stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa., 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.5.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.6 Freedom from foreign matter. When tested as specified in 4.4.3, the fluid path of the syringe and needle shall be free from chips, lint, dirt, or other foreign matter.

3.7 Safety (non-toxicity) and lead content.

3.7.1 Safety (non-toxicity). When tested as specified in 4.4.4, the syringes, including needle protectors, shall comply with safety test requirements of the Pharmacopeia of the United States (USP).

3.7.2 Lead content. When tested as specified in 4.4.4.2 and 4.4.4.3, all plastic and rubber components coming in contact with the medication being administered shall contain not more than 5 parts per million (p.p.m.) of lead.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:54Z
Check the source to verify that this is the current version before use.

3.8 Freedom from pyrogens. When tested as specified in 4.4.12, syringes and needles shall comply with the pyrogen test requirements of the USP.

3.9 Certificates. For destination inspection only, certificates of freedom from pyrogens and safety, as specified below, shall be submitted by the contractor to Defense Personnel Support Center, Directorate of Medical Materiel, ATTN: DPSC-ATQ, 2800 South 20th Street, Philadelphia, Pennsylvania, 19101, and one copy of each shall be forwarded with each shipment to the consignee.

3.9.1 Certificates of safety (non-toxicity). A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety as specified in 3.7.1 and 4.4.4, and were found to comply with such requirements. The certificate shall reference the method used in determining safety.

3.9.2 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens as specified in 3.8 and 4.4.12, and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

3.10 Instructions. Each unit package shall be provided with instructions describing the proper techniques for dispensing the sterile syringes and needles, as applicable, from the containers and preparing them for use. In addition, information as to methods for destroying syringes and needles, as applicable, to prevent their re-use shall also be shown.

3.11 Workmanship. Workmanship shall be first class throughout. Syringes (including components) shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:54Z
Check the source to verify that this is the current version before use.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling for lot acceptance.

4.2.1 Syringes (including needles).

4.2.1.1 Inspection lot. The inspection lot shall be determined in accordance with MIL-STD-105 and, unless otherwise specified, shall be based on the unit of product being one syringe (end item).

4.2.1.2 Sampling for examination. A random sample of syringes shall be selected from each lot in accordance with MIL-STD-105. The inspection level and Acceptance Quality Level (AQL) shall be as indicated in Table III. Unit of product shall be a syringe with a permanently attached needle.

Table III. Sampling for Examination

Type of defect	Inspection level	AQL (percent defective)
Major A	S-4	1.0
Major B	S-1	1.0
Minor A	S-4	4.0
Minor B	S-1	4.0

If acceptance number is not exceeded, lot shall be accepted. If rejection number is equaled or exceeded, inspection at level II may be performed to determine final rejection or acceptance.

4.2.1.3 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table IV.

Table IV. Sampling Criteria for Tests

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective
Barrel	Alcohol Resistance	3.1.2.1	4.4.6	S-1	1.5
Plunger tip	Inertness to medication	3.1.3.1	4.4.7	S-2	0.25
Cannula	Composition	3.1.4	4.4.8	S-1	1.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:54Z
Check the source to verify that this is the current version before use.

Table IV. Sampling Criteria for Tests (cont'd)

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective
Cannula	Corrosion Resistance	3.2.3.1	4.4.13	S-1	1.0
Cannula and barrel	Security of assembly	3.2.3.2	4.4.10	S-1	1.0
Cannula and barrel	Resistance to breakage	3.2.3.3	4.4.11	S-1	1.5
Cannula	Stiffness	3.2.3.4	4.4.9	S-1	1.5
Barrel	Accuracy	3.3	4.4.1	S-1	1.5
Barrel and plunger	Suction test	3.4	4.4.2	S-3	1.5
Syringe	Sterility	3.5	4.4.5	S-3	*
Syringe	Freedom from foreign matter	3.6	4.4.3	S-2	4.0
Syringe	Safety (non-toxicity)	3.7.1	4.4.4	S-1	*
Syringe	Lead	3.7.2	4.4.4.2 & 4.4.4.3	S-1	1.0
Syringe	Freedom from pyrogens	3.8	4.4.12	S-1	*

* Acceptance number shall be zero for all sample sizes.

4.2.1.4 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more characteristics.

4.3 Examination. The syringes shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in Tables III and V. Examination, however, shall not be restricted to defects listed in Table V. Any syringes in the sample containing one or more defects shall be rejected and, if the number of defective syringes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:54Z
Check the source to verify that this is the current version before use.

Table V. Classification of Defects in Accordance with MIL-STD-105

Categories	Defects
Critical	None defined.
Major A	
101	Needle protector missing or dislodged.
102	Needle missing or dislodged.
103	Closed type protector not free of holes, or cotton, or paper seal missing (or inadequate amount) from open type protector.
104	Needle protector frozen to needle or syringe.
105	Plunger rod, plunger tip, or both, missing.
106	Plunger, or plunger tip frozen in barrel.
107	Inside of barrel not coated with required lubricant.
108	Inside of barrel contains excessive lubricant.
109	Graduation markings incomplete, missing, illegible, or incorrect.
110	Barrel not clear, plunger tip not clearly visible making accurate alignment with graduation markings difficult or impossible.
111	Inside surface of barrel not fine uniform surface, free from pits, grind marks, bubbles, mold marks, flash, open air lines, or other defects that would affect fluid tightness and smoothness of its fit with the plunger.
112	Plunger tip not smooth, free from flash, bubbles, breaks, cracks or other defects affecting its fit with plunger.
113	Plunger does not operate smoothly in barrel.
114	Plunger tip detaches from plunger rod when pushed back and forth in syringe barrel.
115	Barrel or plunger (including tip) not free of cracks, breaks, crazing, or tackiness.
116	Immediate container fails to open readily.
117	Immediate container not free from holes, tears, cuts, or punctures.
118	Marking on immediate container missing, not legible, not correct, not complete.
119	Cannula lumen not free of obstructions.
120	Cannula lumen not free of burs, and slivers.
121	Cannula outer surface not smoothly finished or free of pits, tool marks, burs, scale, or corrosion.
122	Point blunt or bent, not sharp, not symmetrical or properly formed, or not free of burs, feathered edges, or other defects (at 10X magnification).
Major B	
151	Barrel wall thickness not as specified.
152	Cannula outside diameter not as specified.
153	Cannula not straight.
154	Cannula inside diameter not as specified.
155	Point length not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:54Z
Check the source to verify that this is the current version before use.

Table V. Classification of Defects in Accordance with MIL-STD-105 (cont'd)

Categories	Defects
Minor A	
201	Dimensions (except barrel wall thickness) not as specified.
202	Deep scratches (not interfering with reading of dosage, see 110) on outside surface of barrel.
203	Contractor's name or registered trademark, missing, incomplete, or illegible.
204	Graduation scale not approximately in line with flattened side of flange.
205	Instructions missing, imcomplete, or illegible.
Minor B	
251	(Unit of product - Unit package and contents) Average count per unit package not as listed in 6.4.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accuracy test. Syringes shall be tested for accuracy at each main graduation mark (see Table I) using water and a calibrated burette. As an alternate to the burette, the temperature and density of the water can be taken into account gravimetrically. The first ring from the distal end of the plunger tip shall be aligned with each graduation mark tested.

4.4.2 Suction test. The fluid tightness test shall consist of the suction test described in the following paragraphs:

4.4.2.1 Suction test. "An amount of water equal to more than one-half the capacity shall be drawn into the syringe through a hypodermic needle. The syringe shall then be held vertically, needle up, to allow air to rise above the water and be expelled. Tapping the barrel of the syringe will facilitate removal of trapped air. The plunger shall be pressed inward to the 50% mark. A soft rubber stopper shall then be used to seal the needle. The plunger shall then be carefully withdrawn, at a rate of about one inch/10 seconds, to the full mark. After the plunger has reached the full mark, there shall be no bubbles coming by the distal ring for at least 10 seconds. Should there be more than three air bubbles passing the distal ring during the 10-second period, there shall be an additional period of observation of 15 seconds during which there shall be no bubbles passing the distal ring. Upon release of the plunger, it shall return to the 50% mark.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:54Z
Check the source to verify that this is the currentversion before use.

Note: Air bubbles which appear on the distal surface of the plunger tip during the withdrawal of the plunger shall be disregarded. Some air bubbles frequently appear under conditions of reduced pressure because of phenomena such as decreased solubility of air in water, interfacial tension, and aggregation of minute bubbles into larger ones. A continuing stream of bubbles arising from the plunger-barrel interface constitutes serious leakage."

4.4.2.2 Alternate suction test. As an alternate, the suction test may be performed with a leak detector which is basically a thermistor conductivity meter using helium. Gas (helium) leakage past the plunger shall be indicated by deflections of the milliammeter needle as listed below for the various sizes of syringes. Suction within the syringe shall be created by withdrawal of the plunger at a rate of approximately one-quarter inch per second. The thermistor conductivity meter which shall be used is the Gow-Mac Model 21-110, or similar apparatus. To minimize hand operation a fixture shall be devised to hold the syringe, couple it via the needle to the apparatus, and work the plunger in a manner coincidental with the center line of the syringe barrel.

Needle Deflection Characteristics (Gow-Mac 21-110)

Size	Sensitivity setting
1 cc	Medium
2-1/2 cc	Medium
3 cc	Medium
5 cc	Low
10 cc	Low
20 cc*	Low
30 cc*	Low

Full scale deflection of the needle (50 divisions) for two or more seconds shall be considered as leakage and cause for rejection.

*Momentary full scale deflection shall not be considered an indication of leakage.

4.4.3 Freedom from foreign matter test. The syringe shall be tested as removed from individual container. The syringe shall be first examined with the naked eye for obvious foreign matter and then shall be submitted to the tests described in the following paragraph:

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:54Z
Check the source to verify that this is the current version before use.

4.4.3.1 Test method. Take an amount of distilled water equal to the capacity of the syringe and pour it into the barrel of the syringe being tested. Insert the piston and eject the distilled water into a medium texture, quantitative filter paper properly inserted into a 60°, 75 mm, smooth glass funnel. As a blank, take an equal amount of distilled water and pour it into a similar, separate funnel filter paper arrangement as used for the syringe. Examine the residue on the filter papers, after the water has passed through them, and determine if foreign material ascribable to the syringe and needle is present.

4.4.4 Safety (non-toxicity) and lead content tests.

4.4.4.1 Safety (non-toxicity). The finished syringes shall be tested for safety in accordance with the USP, for Transfusion and Infusion Assemblies.

4.4.4.2 Test for lead. Ten (10) Gms of rubber or plastic shall be taken, ignited to char, and then further ignited in a muffle furnace at 700° C. to a white ash. The ash shall be boiled in 15 milliliters (ml) of 10 percent nitric acid, filtered into a volumetric flask and made up to 100 ml. An aliquot portion, equivalent to one gram of sample, shall be taken and placed in a separator. The dithizone test for lead, as given in the USP, shall be applied using a control of 5 p.p.m or 12 p.p.m. of lead solution as given in the USP test. If the sample solution is darker than the limit, small aliquots shall be used until the color of the control is matched by the sample solution and the concentration of lead in p.p.m. is determined.

4.4.4.3 Alternate test for lead. Accurately weigh a 5-gram sample into a 1-oz. Vycor crucible. Reduce to ash in a muffle furnace at 650° C for eight hours. Let cool to room temperature and add 15 ml of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50-ml. volumetric flask. Rinse the residue on the filter with deionised water and dilute to 50 ml. Prepare standard lead solutions of 1, 2 and 5 ppm. lead. Determine the absorbances of these solutions at the 283 millimicron wave length using a Hollow Cathode Pb lamp and a 10x scale expansion on a spectrophotometer, such as the Perkin-Elmer Model 303 Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

Calculation:

$$\frac{A \times F \times \text{dilution}}{W} = \text{ppm lead}$$

A = Absorbance

F = Factor = $\frac{\text{(Concentration standard in ppm Pb)*}}{\text{absorbance}}$

W = Sample weight in grams

* The three standards are averaged to give ppm/unit A."

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:54Z
Check the source to verify that this is the current version before use.

4.4.5 Sterility test. The syringe shall be tested for sterility in accordance with the method specified in the USP for parenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.4.5.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in Table IV, Sampling Criteria for Tests.

4.4.5.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65°C
Dry Heat	B. stearothermophilus, or	55° - 65°C
	B. subtilis var. niger (or globigii)	30° - 35°C
Gas	B. subtilis var. niger (or globigii)	30° - 35°C
Radiation	B. pumilus	30° - 35°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.5.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.5 Sterility test and 4.4.5.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.5.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.5.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Request for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:54Z
Check the source to verify that this is the current version before use.

4.4.5.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP.

4.4.6 Permanency of graduation markings. The graduation markings shall withstand five vigorous rubbing cycles of a 70% isopropyl alcohol moistened cotton swab. A cycle shall be one up and one down stroke along the length of the barrel. The markings shall remain legible.

4.4.7 Plunger tip inertness. The rubber plunger tip shall be removed from the plunger rod, wiped with a clean, dry cloth to remove silicone lubricant, and immersed in a test tube under a volume of high pH buffer solution equal to the capacity of the syringe.* The exposure shall be for one hour at 68° to 77° F. (20° to 25° C). There shall be no color or any other visible change of the buffer solution. A control shall be used during the exposure consisting of the same amount of buffer solution in a test tube which does not contain a plunger tip.

* On larger size plunger tips it is permissible to cut them into quarters and use only 1/4 the volume of the buffer solution. The high pH buffer solution shall consist of 50 ml of 0.025 molar borax and 22.7 ml of 0.1 molar sodium hydroxide. The pH of this buffer solution is 10.5.

4.4.8 Chemical analysis. Cannula material shall be subject to chemical analysis in accordance with Methods 111 or 112 of Federal Test Method Std. No. 151.

4.4.9 Stiffness test. The cannula shall be placed across two test edges (0.002 inch radius maximum) separated by 0.375 inch $\pm$ 0.003 inch. A 25-ounce $\pm$ 1/4 ounce load shall be applied at the center of the span and the resulting maximum deflection of the cannula shall be measured using a gage with graduations of at least 0.001 inch. The force shall be applied through an edge with a maximum radius of 0.036 inch, and centered to within $\pm$ 0.003 inch.

4.4.10 Security of assembly. A straight pull of five pounds shall be applied between the cannula and barrel for a period of one minute.

4.4.11 Resistance to breakage test. The barrel shall be held in a fixed position and a force shall be applied to the cannula at a 5/16 inch distance from the barrel tip so that the cannula can be bent clockwise through an arc of 25 degrees from the original position. The operation shall then be reversed (i.e., bent through an arc of 25 degrees from the neutral position in a counter-clockwise direction), and finally the cannula shall be returned to the neutral position. This complete operation (or cycle) shall be repeated 20 times.

4.4.12 Freedom from pyrogens test. The pyrogen test shall be performed as described in the USP, for Transfusion and Infusion Assemblies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:54Z
Check the source to verify that this is the current version before use.

4.4.13 Corrosion-resistance of cannula. The cannula shall be immersed in 10 percent citric acid at room temperature for 5 hours and, after removal from the acid, shall be boiled in distilled water for 30 minutes and, then shall remain immersed for 48 hours in the distilled water. At the end of the 48-hour period, the cannula shall be allowed to dry by evaporation, and shall then be examined for evidence of corrosion. A glass vessel shall be used throughout this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package containing twenty-five syringes, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table VI shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table VI. Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	40 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Twenty-five sterile disposable syringes shall be packaged in a premolded plastic container. The container shall consist of two nested set-up trays so designed that each syringe is firmly held within an individual compartment. The compartment shall be designed for ease in removal of the syringe. Closure of the container shall be with a sealed peel open plastic cover. The sealed container shall be capable of maintaining sterility of contents unless damaged or opened.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:54Z
Check the source to verify that this is the current version before use.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Fiberboard boxes shall conform to special requirements specified in Table II of PPP-B-636.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-601, domestic type; PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636. In addition, fiberboard boxes shall conform to the special requirements specified in Table 1 of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. In addition to any commercial markings, each unit package shall be marked in accordance with MIL-STD-129. Sterility lot (control) number, needle and syringe size, name and address of contractor and contract or purchase order number. The method of sterilization (G-Gas, R-Radiation, C-Cold, S-Steam, D-Dry Heat Sterilization) shall be shown following the level of packaging and date of packing. Example: "A-7/74-G". Final sterilization shall be on or after indicated packaging date. In addition, marking shall include the legends that syringes are sterile, pyrogen free, disposable and "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED" or similar statement

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:54Z
Check the source to verify that this is the current version before use.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number and the method of sterilization as specified in 5.3.1 shall be shown.

NOTE: Each container of 40 unit packages shall include one copy of the following advisory notice either inside or on the outside of the package:

"CAUTION: RECOMMEND THAT MEDICATION WHICH IS LEFT IN DISPOSABLE SYRINGES FOR MORE THAN 15 MINUTES BE EXAMINED, PRIOR TO ADMINISTRATION, FOR ANY CHANGE IN APPEARANCE."

6. NOTES

6.1 The syringe specified herein is part of an allergist therapy unit of 25 sterile, disposable syringes of 1 cc capacity and permanently affixed needles.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and title of this specification.
(b) Applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification covers the following item as listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-001-7680	SYRINGE AND NEEDLE, HYPODERMIC, DISPOSABLE, Allergy, 26 ga, 3/8 inch Intradermal Needle, 1 cc, 25s

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1849

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-S-37072

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:54Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, Pa. 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:54Z
Check the source to verify that this is the current version before use.